

Sonia Nicoletti

PhD Student in Computer Science

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Stuttgart, Germany

EDUCATION

PhD in Computer Science

Max Planck Institute for Intelligent Systems - CS@max planck program

Stuttgart, Germany
Sep 2025 – Ongoing

- *Supervisors:* Ksenia Keplinger, Philipp Müller
- *Research areas:* Human-AI Interaction, LLMs, NLP, Computational Social Science

Master's Degree in Computer Science

University of Bologna

Bologna, Italy
Sep 2022 – Dec 2024

- *Final grade:* 110/110 cum laude
- *Thesis:* "LLMs and Essence: Developing a Chatbot to Support Software Engineering Practices"
- *Study abroad:* National Taiwan University (Taiwan, Sep 2023 – Dec 2023)
- *Study abroad:* Drexel University (USA, Sep 2022 – Dec 2022)

Bachelor's Degree in Computer Science for Management

University of Bologna

Bologna, Italy
Sep 2019 – Jul 2022

- *Final grade:* 110/110 cum laude

RESEARCH AND TEACHING EXPERIENCE

Tutor & Research Assistant

University of Bologna, supervised by Paolo Ciancarini

Bologna, Italy
Mar 2025 – Aug 2025

- Designed, implemented, and deployed an **end-to-end** LLM-powered chatbot to support software engineering education and practice.
- Built a custom **Retrieval-Augmented Generation** pipeline, including document ingestion, embedding, and **hybrid retrieval** (vector search with cosine similarity + BM25), which improved answer accuracy by 28%.
- Delivered a lecture to GEPID's master students and co-authored instructional material on software engineering practices and **Agile methodologies**; supported the evaluation of student projects and supervised a student's thesis.

PROJECTS

AI-delivered Bias Training

Organizational Leadership and Diversity (MPI-IS), supervised by Ksenia Keplinger

Nov 2025 – Ongoing

- Designed an online lab experiment measuring **leadership evaluations** and participants' **helping behavior** (time allocated and effort) before and after a 15-minute unconscious bias training delivered by an **AI avatar**.
- The study extends a prior field experiment in a multinational fintech company to assess whether AI-delivered training **reduces nationality-based biases** in leadership perceptions as effectively as in-person training.

Walking the DEI Talk

Organizational Leadership and Diversity (MPI-IS), supervised by Ksenia Keplinger

Nov 2025 – Ongoing

- Developed a dual-study research project examining **CEO communication on DEI**, integrating motivation theory with computational text analysis, to distinguish genuine commitment from performative discourse and linking patterns to organizational outcomes (e.g., LGBTQ+ inclusion ratings, board diversity, minority representation).
- Designed a **weakly supervised NLP classifier** using large language models to extract authenticity signals from DEI segments of CEOs' interviews (collected from YouTube) and quantify them through theory-driven linguistic markers.

Spook the Machine

Center for Humans and Machines (MPI-HD), supervised by Levin Brinkmann

Nov 2025 – Mar 2026

- Designed and deployed a **multi-agent affective system** of 102 LLM-powered chatbots with personalised phobia profiles and reward mechanisms, collecting data from 800+ users and ~15,000 human-machine interactions.
- Analysed **collective exploration dynamics** across user-generated prompts using BERT-based semantic clustering, Minimum Spanning Tree weight over time, and conditional log-likelihood measures.
- Conducted sentiment analysis on LLM-generated fear responses across phobia types and emotional conditions to evaluate the influence of affective architectural design choices on **machine emotional expression**.

PUBLICATIONS

- Sonia Nicoletti and Paolo Ciancarini. 2026. "Essence Coach: A Bot for Software Practice Adoption". In *Proceedings of the 2026 7th International Workshop on Bots and Agents in Software Engineering (BoatSE '26)*. Association for Computing Machinery, New York, NY, USA, 27–34. <https://doi.org/10.1145/3786161.3788464>

TALKS

- “Empowering Software Teams with Essence: An LLM-Driven Approach to Supporting Practice Adoption” – *Essence Education Forum*, hosted by Prof. Stéphane Gagnon, University of Quebec in Outaouais, Canada (Virtual), December 2025.

HONORS AND AWARDS

- The Cornell, Maryland, Max Planck Pre-doctoral Research School**
Program attendee

2023
- Selected as one of the ~90 students to attend the Cornell, Maryland, Max Planck Pre-doctoral Research Summer School.

- UNIBO Study Award for Computer Science**
Scholarship recipient

2023
- Selected as one of two recipients in the Computer Science master’s program based on academic performance.

PROFESSIONAL EXPERIENCE

- Full-Stack Web Developer**
Cooki srl

Bologna, Italy

Jan 2022 – Mar 2022
- Refactored the **relational database** schema (ER model) and optimised SQL queries on user-related tables, reducing query response time by 40%.
 - Developed a **JavaScript** feature to generate food labels for 20,000 products from database data dynamically.
 - Conducted requirements analysis and designed 10+ **UI mockups** for a mobile application using Figma.

SKILLS

Front-end: HTML/CSS, JavaScript (Vue, React), Kotlin
Back-end: Java (JUnit), Node.js (Express), SQL (MySQL, Postgres), Firebase, MongoDB
Data science and AI/ML: Python (PyTorch), R
Developer Tools: Git, Figma, Jira, Agile (Scrum)

LANGUAGES

English (C2), Italian (native), French (C1)